

Exploratory Data Analysis Report

Python + SQL Project

(ITSM Service Desk Operations Analytics Dataset)

EDA report based on the completed Data Cleaning, Univariate, Bivariate and Multivariate analysis work.

Dataset	ITSM Ticket and SLA Data
Analysis records	360,000
Analysis columns	29
Analysis period	April 2021 – July 2021
Tools used	Python, Pandas, NumPy, Matplotlib

1. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure, quality and important patterns in the ITSM service desk data. The analysis mainly focuses on ticket activity, SLA performance, actual handling time and customer satisfaction.

The analysis was divided into three parts:

- **Univariate Analysis** – to study individual variables.
- **Bivariate Analysis** – to study relationships between two variables.
- **Multivariate Analysis** – to study multiple variables together.

The purpose of the EDA was not only to create charts, but to identify patterns that can be connected to service performance and SLA management.

2. Dataset Used for EDA

The analysis used the cleaned dataset created after loading the ticket and SLA information from PostgreSQL and merging the two DataFrames in Jupyter Notebook using Pandas.

Item	Result
Original ticket data	180,000 records × 20 columns
SLA data	360,000 records × 8 columns
Merged analysis data	360,000 records × 27 columns
Final cleaned data	360,000 records × 29 columns
Derived columns	2
Derived fields	sla_variance_minutes, sla_performance_percentage

The SLA table contains two SLA records per ticket: **First Response** and **Resolution**. This explains why the merged dataset contains 360,000 rows even though there are 180,000 tickets.

3. Data Quality and Preparation

Before analysis, the merged data was checked for missing values, duplicate rows, invalid values and potential numerical outliers.

Check	Result	Interpretation
Duplicate rows	0	No exact duplicate rows were found.
Duplicate ticket IDs	180,000	Expected because each ticket has two SLA records.
Missing CSAT values	61,446	Customer feedback was not available for these records.
Invalid CSAT ratings	0	Available ratings were within 1–5.
Negative SLA actual time	0	No negative values found.
Negative SLA target	0	No negative values found.
Negative breach minutes	0	No negative values found.
Potential numerical outliers	0	IQR check returned zero potential outliers.

Two derived columns were created for further analysis.

- **sla_variance_minutes** measures actual time minus the SLA target.
- **sla_performance_percentage** expresses actual time as a percentage of the SLA target.

4. Univariate Analysis

Univariate analysis was used to understand the distribution of individual numerical and categorical variables. The analysis included statistical summaries, customer satisfaction distribution and the two derived SLA performance measures.

Key Observations

- The average customer satisfaction rating was approximately **3.01 out of 5** for records where a rating was available.
- The customer satisfaction ratings were distributed across all values from **1 to 5**, while **61,446 ratings were missing**.
- Average actual SLA time was approximately **3,053.57 minutes** across the merged SLA records.
- Average breach time was approximately **2,281.56 minutes**.

- Average SLA variance was approximately **2,273.57 minutes**.
- Average SLA performance percentage was approximately **293.84%**, showing that the merged SLA records were, overall, well above the target percentage when actual time is compared with the target.

Recommended Visuals for the EDA PDF

Add:

- Customer Satisfaction Rating Distribution
- Distribution of SLA Variance
- Distribution of SLA Performance Percentage

Leave Out:

- Histograms for ID columns such as ticket_id and sla_log_id because they do not provide useful business insight.

5. Bivariate Analysis

Bivariate analysis was used to study relationships between SLA variables, resolution codes, customer satisfaction and service performance.

Important Data Limitation

Several categorical fields contained only one value in the available data.

For example, all records were:

- Incident tickets
- P3 priority
- Medium impact
- Medium urgency
- Closed status
- Email source

Therefore, comparisons such as **Priority vs Ticket Status** do not provide meaningful variation and should not be treated as business findings.

Key Findings

- Resolution code had **eight categories**, with fairly similar record counts. This made it suitable for comparison with SLA status.
- SLA status showed a strong difference in actual time: **breached records averaged about 3,535.93 minutes**, while **met records averaged about 62.50 minutes**.
- Average SLA variance was about **+2,649.50 minutes for Breached records** and **-57.50 minutes for Met records**.
- For SLA type, **First Response** contained both Met and Breached records, while **all Resolution SLA records were Breached**.
- The **CSAT vs Actual Time** scatter plot did not show an obvious strong relationship between actual handling time and customer satisfaction.

Recommended Visuals

- Add: SLA Status by Resolution Code
- Add: Actual Time vs SLA Target Time
- Add: Actual Time by SLA Status
- Add: Breach Minutes by SLA Status
- Add: Customer Satisfaction vs Actual Time
- Add: SLA Status by SLA Type
- Add: Average SLA Variance by SLA Status
- Optional: Actual Time by Resolution Code, because the differences between resolution-code averages are small.

6. Multivariate Analysis

Multivariate analysis combined SLA type, SLA status, actual time, SLA variance, SLA performance percentage and customer satisfaction to identify broader patterns.

Key Findings

- For **First Response SLAs**, breached records averaged **272.46 minutes** and **227.05% SLA performance**, while met records averaged **62.50 minutes** and **52.08%**.

- **Resolution SLAs** averaged **5,893 minutes** and **409.24% SLA performance**, and no Resolution SLA record was marked Met.
- When grouped by SLA status, breached records had an average CSAT of about **3.01**, actual time of about **3,535.03 minutes** and SLA performance of about **332.82%**.
- Met records had an average CSAT of about **3.01**, actual time of about **62.50 minutes** and SLA performance of about **52.08%**.
- By SLA type, **First Response** averaged **214.14 minutes actual time** and **94.14 minutes SLA variance**, while **Resolution** averaged **5,893 minutes actual time** and **4,453 minutes SLA variance**.

Recommended Visuals

- Add: SLA Performance by SLA Type and SLA Status
- Add: Multivariate Comparison of CSAT, Actual Time and SLA Performance by SLA Status
- Add: SLA Metrics by SLA Type
- Leave out: the resolution-code multivariate graph because the differences between resolution-code groups are very small.
- Several very high correlations come from mathematically related SLA measures.

7. Overall Key Insights from EDA

1. SLA breaches are the main performance concern

The analysis shows a very large difference between Met and Breached SLA records. Breached records have much higher actual time and positive SLA variance.

2. Resolution SLA performance requires attention

All **180,000 Resolution SLA records** in the analysed data are marked Breached. This is the clearest SLA-related issue in the dataset.

3. First Response performance is comparatively better

First Response records contain both Met and Breached outcomes, and the average actual time for Met records is much lower than for Breached records.

4. Customer satisfaction is relatively stable across SLA status

Average CSAT is approximately **3.01 for both Breached and Met records**. The available analysis does not show a strong difference in average satisfaction by SLA status.

5. Resolution code does not appear to explain the major time difference

Average actual time is broadly similar across the eight resolution codes. Therefore, resolution code alone does not explain the large SLA performance gap.

6. The dataset has limited categorical variation

Many ticket-level categorical fields contain only one category. This limits the usefulness of some standard bivariate comparisons and should be documented as a data limitation.

7. Derived SLA columns improved the analysis

`sla_variance_minutes` and `sla_performance_percentage` provided additional measures that made SLA performance easier to compare across groups.

8. Business Interpretation

From a service-management perspective, the main issue identified through EDA is SLA performance rather than ticket categorization. The strongest pattern is the difference between SLA records that were met and those that were breached.

- Management should investigate why Resolution SLAs are consistently marked as breached.
- First Response performance can be used as a comparison point because it contains both successful and breached SLA outcomes.
- Large positive SLA variance indicates that actual handling time is substantially above the defined target for breached records.
- Customer satisfaction should still be monitored, but the current EDA does not establish a strong direct relationship between satisfaction and actual handling time.
- Future analysis would benefit from more variation in priority, category, ticket type, source channel and ticket status.

9. Limitations and Data Notes

- The available ticket-level categorical variables have very limited variation, with several fields containing only one category.
- The merged dataset contains two SLA records per ticket, so row counts should not be interpreted as unique ticket counts.
- Missing customer satisfaction ratings were retained because they represent unavailable feedback rather than invalid data.
- The correlation matrix contains strong relationships between variables that are mathematically derived from the same SLA measures. These should not be interpreted as independent business drivers.
- The EDA findings are based only on the available project dataset and should not be generalized beyond this data without further validation.

10. Conclusion

The EDA provided a clear view of the quality and SLA behaviour of the ITSM dataset. The data was successfully validated and prepared, and two useful derived SLA measures were created for deeper analysis.

The strongest finding is the gap between SLA targets and actual handling time for breached records, especially for Resolution SLAs. At the same time, customer satisfaction remained close to **3.01 on average for both SLA statuses**, so the current analysis does not show a strong direct relationship between SLA status and average CSAT.

Overall, the analysis indicates that SLA management and resolution performance should be the main focus for further investigation. The limitations in categorical variation should also be considered when interpreting the results.

THANKYOU!!!!
